OFFICIAL CURRICULUM PROPOSAL

# Artificial Intelligence & Machine Learning

From Pure Basics to Production Pro with End-to-End AI Projects

| TARGET TRACK | METHODOLOGY | OUTCOME |
| --- | --- | --- |
| Pure Beginners to AI Engineers | First Principles + Production Labs | Enterprise Portfolio Ready |

PROGRAM OVERVIEW

# Connecting Core Foundations to Modern Generative AI

### Pure First Principles

Build true mathematical and code intuition from Scratch—no black-box magic, starting from basic Python and linear algebra.

### Deep Learning & GenAI

Master modern PyTorch, Computer Vision, Transformers, LLM Fine-Tuning, and RAG systems using vector databases.

### Production Deployment

Containerize AI models using Docker, build robust REST APIs with FastAPI, and deploy scalable ML services.

LEARNING OUTCOMES

# Core Engineering Competencies Developed

### Applied AI Mathematics

Master Linear Algebra, Vector Calculus, Optimization (Gradient Descent), and Probability required for deep modeling.

### Classical ML Pipelines

Implement Regression, Classification, Clustering, and Gradient Boosted Trees (XGBoost/LightGBM) with Scikit-Learn.

### Neural Networks & PyTorch

Design, train, and debug Deep Neural Networks, CNNs, RNNs, and Attention mechanisms from scratch in PyTorch.

### Generative AI & MLOps

Build production RAG pipelines, fine-tune LLMs, and containerize models with Docker & FastAPI for real-world hosting.

AUDIENCE & PREREQUISITES

# Who Is This Course Designed For?

Target Audience

## Candidate Profiles

- ✓ CS / STEM Students moving into AI research or engineering
- ✓ Software Engineers pivoting into Machine Learning roles
- ✓ Data Analysts wanting to move beyond traditional BI
- ✓ Motivated beginners looking for a complete end-to-end roadmap

Entry Requirements

## Zero High-Level Prerequisites

- › Basic high school algebra and logical problem-solving ability
- › General computer operations and software literacy
- › No prior Python or Machine Learning knowledge strictly required
- › Strong dedication to active hands-on coding labs

CURRICULUM ARCHITECTURE

# 5 – Stage Master Roadmap

STAGE 01

## Basics & Math

Python, NumPy, Linear Algebra, Calculus, Matrices.

→

STAGE 02

## Classical ML

Regression, Trees, XGBoost, Clustering, Evaluation.

→

STAGE 03

## Deep Learning

PyTorch, Neural Networks, CNNs, Vision Models.

→

STAGE 04

## Generative AI

Transformers, Hugging Face, RAG & Vector Databases.

→

STAGE 05

## Pro MLOps

FastAPI, Docker, Cloud Hosting & CI/CD Deployment.

FOUNDATIONS TRACK

# Stage 1 & 2: Foundations, Math & Classical ML

## Stage 1: AI Math & Python

- Python OOP, Data Structures, and Vectorized operations in NumPy
- Linear Algebra: Vectors, Matrices, Eigenvalues, Vector Spaces
- Multivariate Calculus: Partial Derivatives & Gradient Descent intuition
- **Project 1:** Writing a Gradient Descent Engine from Scratch in Python

## Stage 2: Classical Machine Learning

- Supervised Learning: Linear/Logistic Regression, Decision Trees
- Ensemble Power: Random Forests, XGBoost, and LightGBM
- Unsupervised Learning: K-Means Clustering, PCA Dimensionality Reduction
- **Project 2:** End-to-End Enterprise Customer Churn & Pricing Predictor

DEEP LEARNING TRACK

# Stage 3: Deep Neural Networks & Computer Vision

## Neural Network Mechanics

- Deep Learning Foundations with PyTorch: Tensors, Autograd, Optimizers
- Multi-Layer Perceptrons (MLPs) & Custom Loss Functions
- Computer Vision: Convolutional Neural Networks (CNNs) & ResNet Transfer Learning
- Regularization Techniques: Dropout, Batch Normalization, Data Augmentation

## End-to-End Project 3

**Real-Time Medical Image Diagnostic Engine:**
Building a fine-tuned Computer Vision classifier in PyTorch to detect medical anomalies from X-Ray/MRI imagery with automated heatmaps using Grad-CAM.

MODERN AI TRACK

# Stage 4: Generative AI, Transformers & LLMs

## Generative AI Architectures

- Sequence Modeling & Attention Mechanisms (Self-Attention, Multi-Head)
- The Transformer Architecture & Hugging Face Ecosystem
- Large Language Models (LLMs) & Parameter Efficient Fine-Tuning (LoRA, PEFT)
- Vector Databases (Pinecone/ChromaDB) & Semantic Search RAG Systems

## End-to-End Project 4

**Enterprise Knowledge Base Copilot (RAG System):**
Building a production-ready Generative AI assistant integrating custom PDF datasets, vector embeddings, re-ranking models, and an interactive interface.

PRO TRACK & MLOPS

# Stage 5: Pro Level MLOps & Cloud Deployment

## Production Engineering

- Packaging AI models into high-performance REST APIs with FastAPI
- Containerizing model environments using Docker & Docker Compose
- Experiment Tracking, Lineage, and Model Registry with MLflow
- Continuous Integration & Continuous Deployment (CI/CD) pipelines

## Final Capstone Project

**Full-Stack Cloud AI Microservice:**
Deploying a trained PyTorch / Transformer model packaged inside a Docker container onto AWS/GCP, monitored with MLflow and accessible via a production API.

PEDAGOGY & DELIVERY

# The Active Learning & Production Lab System

### 1. Code-First Math

Mathematical theory is introduced alongside live Jupyter notebook execution for immediate intuitive understanding.

### 2. Code Reviews

Weekly GitHub PR reviews by experienced senior AI engineers enforcing clean modular code and efficient architecture.

### 3. Industrial Sprints

Simulated sprint environments where students tackle messy enterprise datasets and deploy functional prototypes.

EVALUATION FRAMEWORK

# Multi-Tier Student Assessment Strategy

## 40%

### End-to-End Projects

4 major milestone projects evaluated on code quality, model accuracy, and modular structure.

## 30%

### Coding Labs & Quizzes

Weekly Python, PyTorch, and optimization challenges testing execution speed and logic.

## 30%

### Capstone Cloud Defense

Live technical presentation and architectural defense of the deployed cloud AI microservice.

READY FOR LAUNCH

# Empower Students with Production AI Capabilities

Complete syllabus, PyTorch notebooks, production labs, and MLOps deployment frameworks ready for immediate integration.

Accept Course Proposal